

2016.0 RANGE ROVER (LG), 501-29

SIDE PANEL SHEET METAL REPAIRS

ROCKER PANEL SECTION

[G 1 4 5 4 6 3 3]

REMOVAL AND INSTALLATION

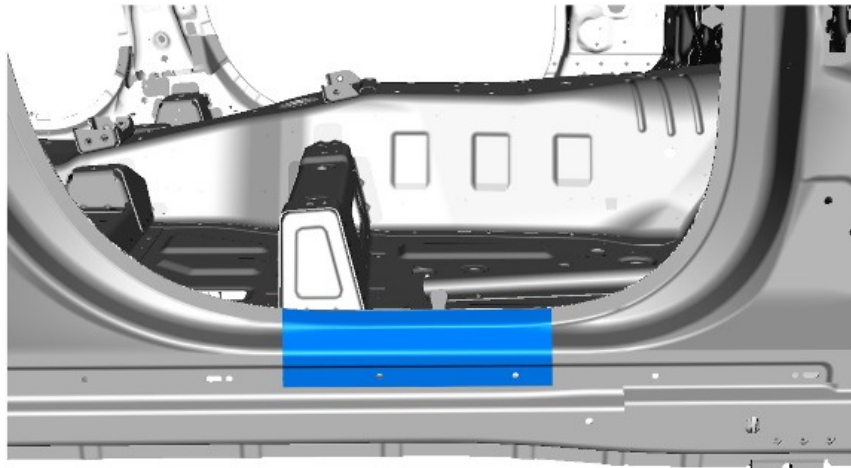
REMOVAL

NOTE:

The rocker panel section is installed in conjunction with:

- Rear scuff plate trim panel
 - Rocker panel moulding
 - Front seat
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1.



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The rocker panel section is serviced as indicated.

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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

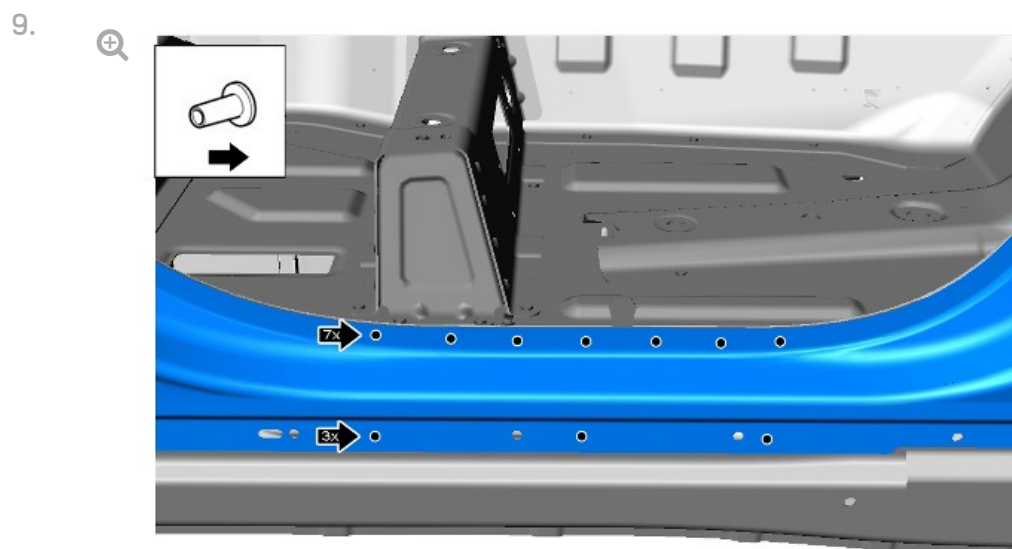
For additional information, refer to: Health and Safety Precautions (100-00 General Information, Description and Operation).

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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

4. For additional information, refer to: Rear Rocker Panel Moulding (501-08 Exterior Trim and Ornamentation, Removal and Installation).
5. For additional information, refer to: Rear Scuff Plate Trim Panel (501-05 Interior Trim and Ornamentation, Removal and Installation).
6. For additional information, refer to: Front Seat (501-10 Seating, Removal and Installation).
7. Release the floor carpet and position it to one side.
8. Release the rocker panel wiring harness and position to one side.



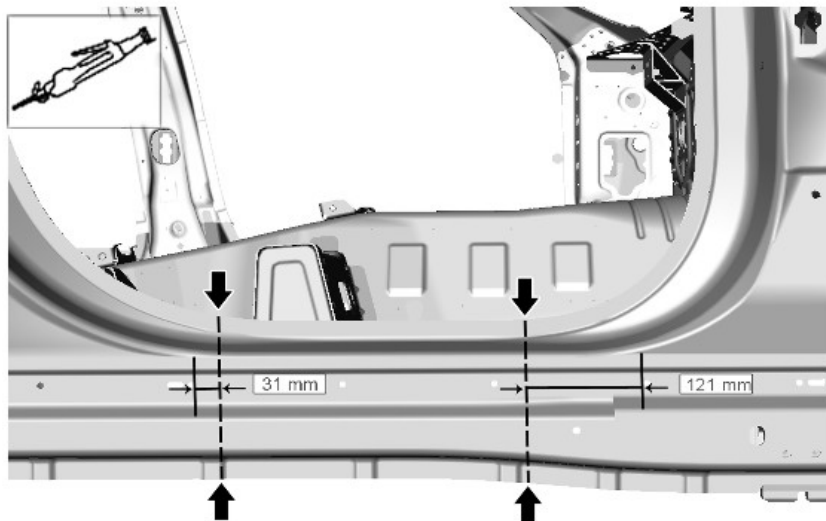
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Remove the SPR's as indicated.

10.

NOTE:

Care should be taken not to cut through into inner panels or reinforcements.



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Mark out the positions where the rocker panel section MIG (metal inert gas) butt joint are to be made. Cut through the rocker panel at these points as indicated.

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11. Separate the joints and remove the rocker panel section.
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INSTALLATION

CAUTION:

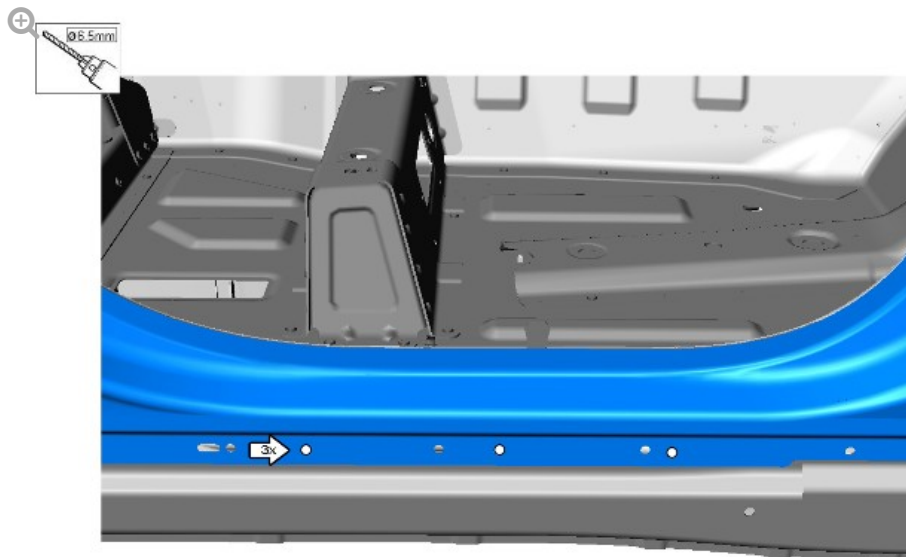
Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

1. Remove rivet remnants.
2. Dress flanges where necessary.
3. Using a fine bristle disc, clean and prepare the panel surfaces.
4. Mark out the positions on the new rocker panel section where the MIG butt joint are to be made and cut through the rocker panel at these points.
5. Offer up the new rocker panel section and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.

6.

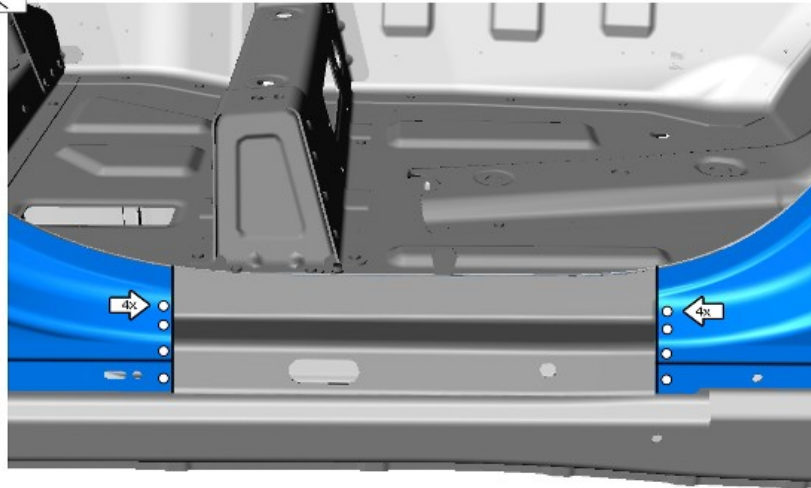


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Drill holes through the old rocker panel section into the new rocker panel section where BSF's are to be installed as indicated.

7. Remove the new rocker panel section.

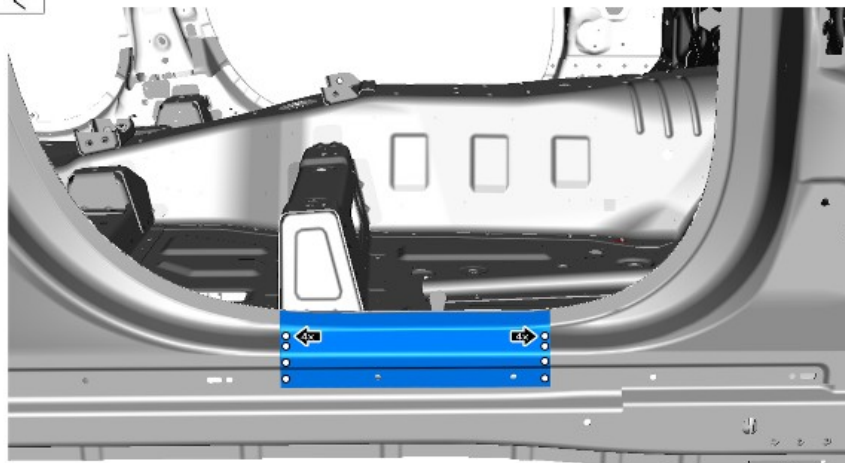
8.



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Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

9.



E143384

Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

10. Deburr the drilled holes.

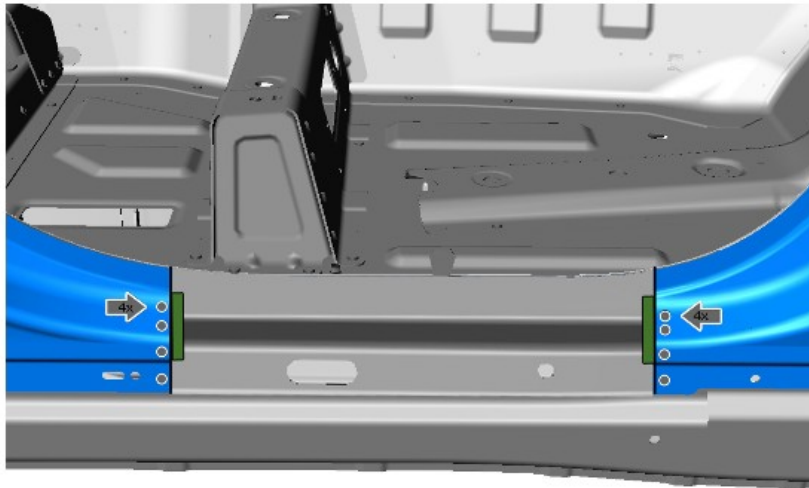
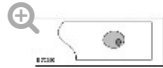
11. Measure and cut two 50mm wide backing plates, 25mm each side

of the MIG butt joint, from the new rocker panel remnant.

12. Cut four run on/run off tab from the new rocker panel remnant.

13. Using a fine bristle disc, clean and prepare the panel surfaces.

14.



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Install and MIG plug weld the backing plates as indicated.

15. Apply the coupling agent where the Land Rover approved bonding material is to be applied and allow to dry.

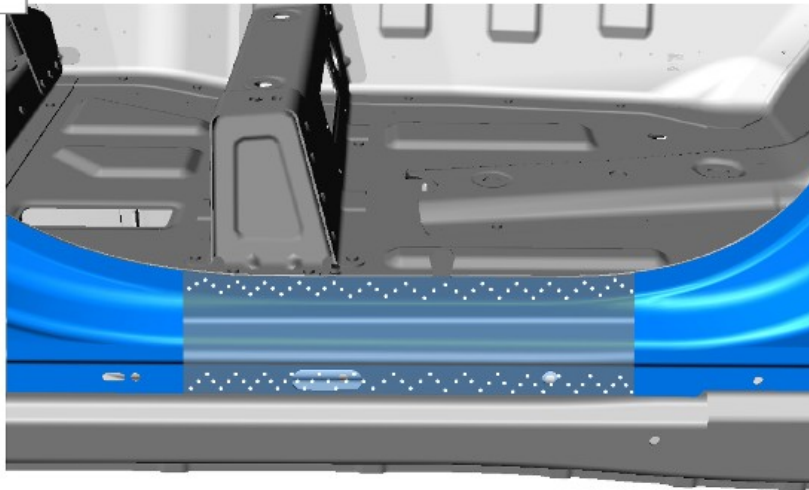
16.

CAUTION:

Do not apply the Land Rover approved bonding material in the vicinity of the MIG welds.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.

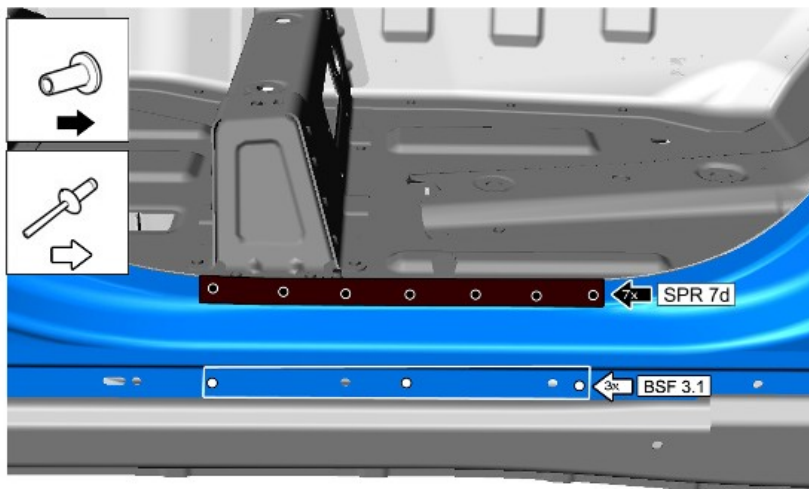


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Apply a 5mm zig zag bead of the Land Rover approved bonding material as indicated.

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17. Offer up the new rocker panel section and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
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18.



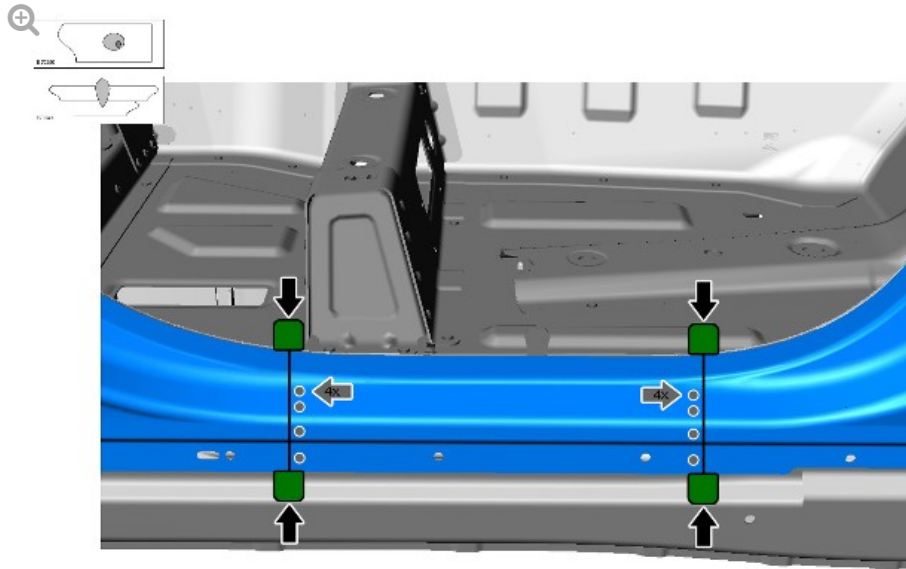
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Install the SPR's and the BSF's as indicated.

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19. Remove any excess adhesive.

20. Install and tack weld the run-on/run-off tabs to the rocker panel.

21.



E 143388

MIG plug weld the backing plates and MIG weld the rocker panel section butt joints as indicated.

22. Cut off the run on/run off tabs.

23. Carry out a non destructive crack inspection on the MIG butt joints. If correct proceed to next step, if not, rectify and recheck before proceeding.

24. Dress the welded joints.

25. Make sure that any open or exposed panel joints are suitably sealed.

26. Make sure corrosion protection is applied to all areas affected by repair.

27. The installation of associated panels and components is the reversal of removal procedure.
